Assignment 3 :DataFrame

Dataset: Consider the following dataset of sales records:

Order ID	Customer	Product	Category	Quantity	Price	Total Sales
1001	Alice	Laptop	Electronics	2	800	1600
1002	Bob	Phone	Electronics	1	500	500
1003	Charlie	Chair	Furniture	4	100	400
1004	Alice	Table	Furniture	1	250	250
1005	David	Laptop	Electronics	1	800	800
1006	Charlie	Phone	Electronics	2	500	1000
1007	Bob	Chair	Furniture	2	100	200
1008	Alice	Phone	Electronics	1	500	500

Tasks:

1. **Create a DataFrame:**
   - **Load the above dataset into a Pandas DataFrame.**
2. **Add a New Column:**
   - **Add a column Discount that gives a 10% discount on total sales.**
3. **Add a New Row:**
   - **Add a new sales record: Order ID = 1009, Customer = "Eve", Product = "Tablet", Category = "Electronics", Quantity = 1, Price = 300, Total Sales = 300.**
4. **Filter Data:**
   - **Retrieve only the sales records where the Category is "Electronics".**
5. **Selection:**
   - **Select the Customer and Total Sales columns.**

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6. **Grouping:**
    - **Group the data by Customer and calculate the total sales per customer.**
7. **Sorting:**
    - **Sort the DataFrame based on Total Sales in descending order.**
8. **Apply Function:**
    - **Apply a lambda function to create a new column Sales Category, where values are labeled as "High" if Total Sales > 1000, otherwise "Low".**
9. **Save DataFrame:**
    - **Save the final DataFrame as a CSV file named sales_data.csv.**
